COVID-19, Schiphol and vacations abroad

Project group 11 · TIL6022 Python Programming · Version 5 · 3 November 2026

> **Key finding.** COVID-19 almost completely wiped out passenger traffic at Schiphol in April 2020, and Dutch people also genuinely went on fewer vacations abroad during that period. Flights and vacations fell and recovered together: they moved in the same direction in 7 of the 8 years, with a correlation of 0.977. In 2025 flights were back at 96% of 2019, while the number of Dutch people vacationing abroad was at 87%. These figures do not establish cause and effect.

1. What happened at Schiphol

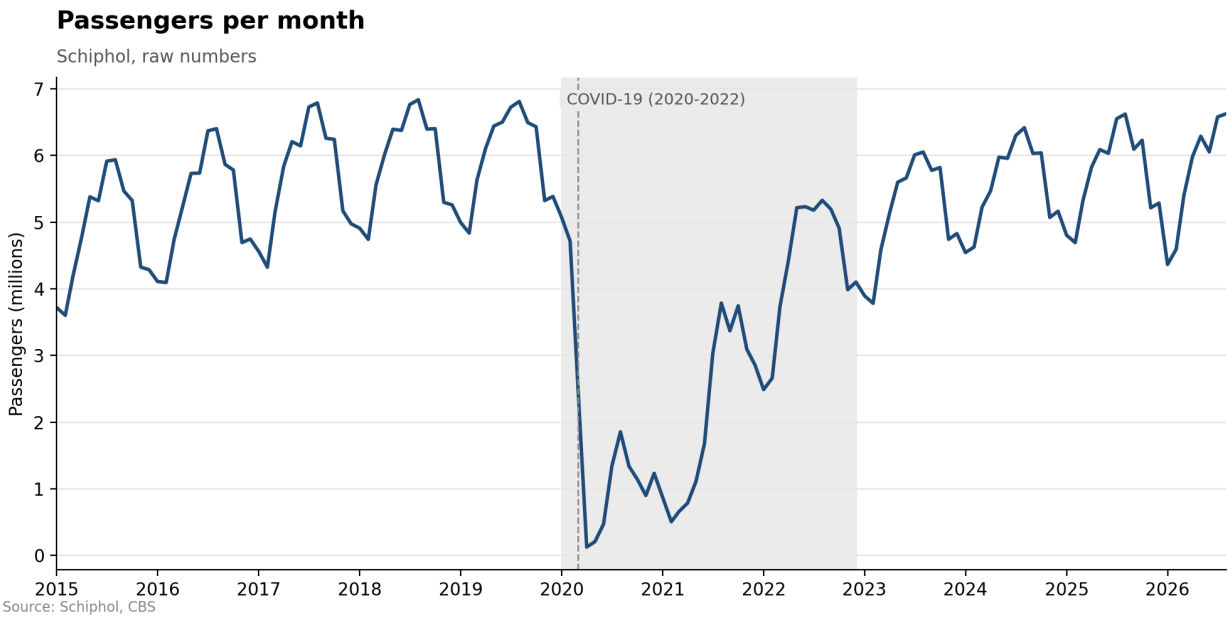


Figure 1. Monthly passenger numbers at Schiphol, without any correction.

Before COVID-19, Schiphol handled between roughly 3.6 and 6.8 million passengers a month, with a clear summer peak every year (the highest month was August 2018, at 6.84 million). In April 2020 this fell to 126,015 passengers, compared with 6,104,010 in April 2019: a drop of 97.9%. Since 2023 the familiar seasonal pattern has returned at close to the old level.

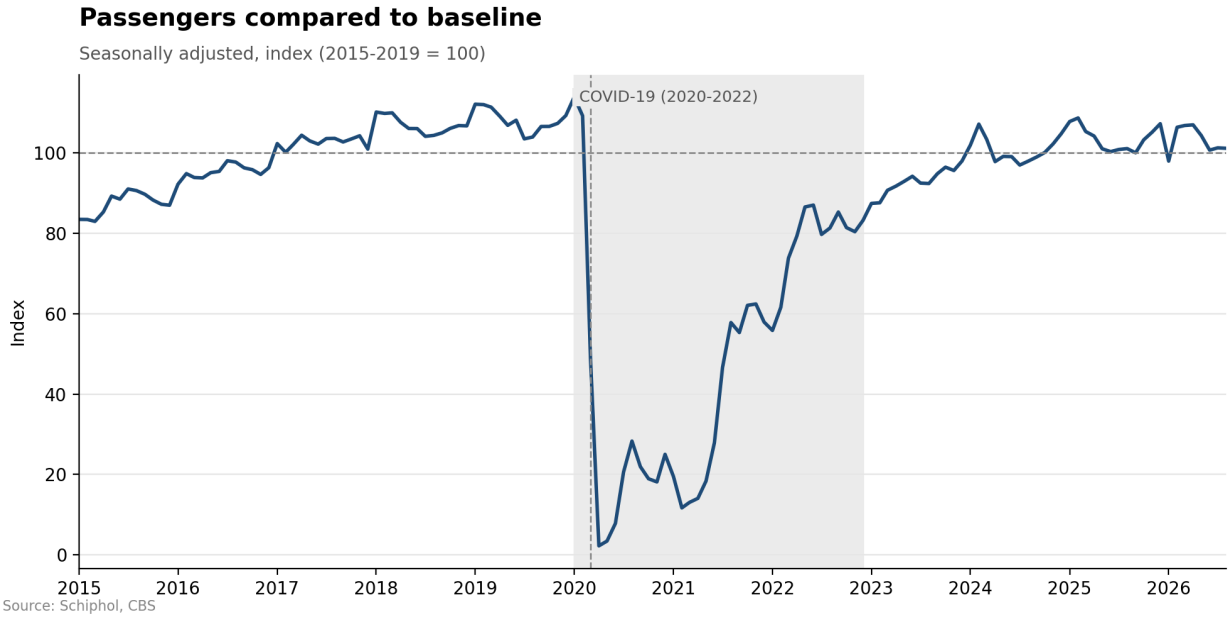


Figure 2. Each month is compared with the same calendar month in 2015-2019, which removes the seasonal pattern.

With the seasonal pattern removed, the shock and the recovery are easier to see: the index fell to 2.3 in April 2020 and only climbed back steadily from mid-2021. Schiphol first rose above the 2015-2019 average again in January 2024, and has been at or above 100 in almost every month since October 2024 (the exception is January 2026, at 97.9).

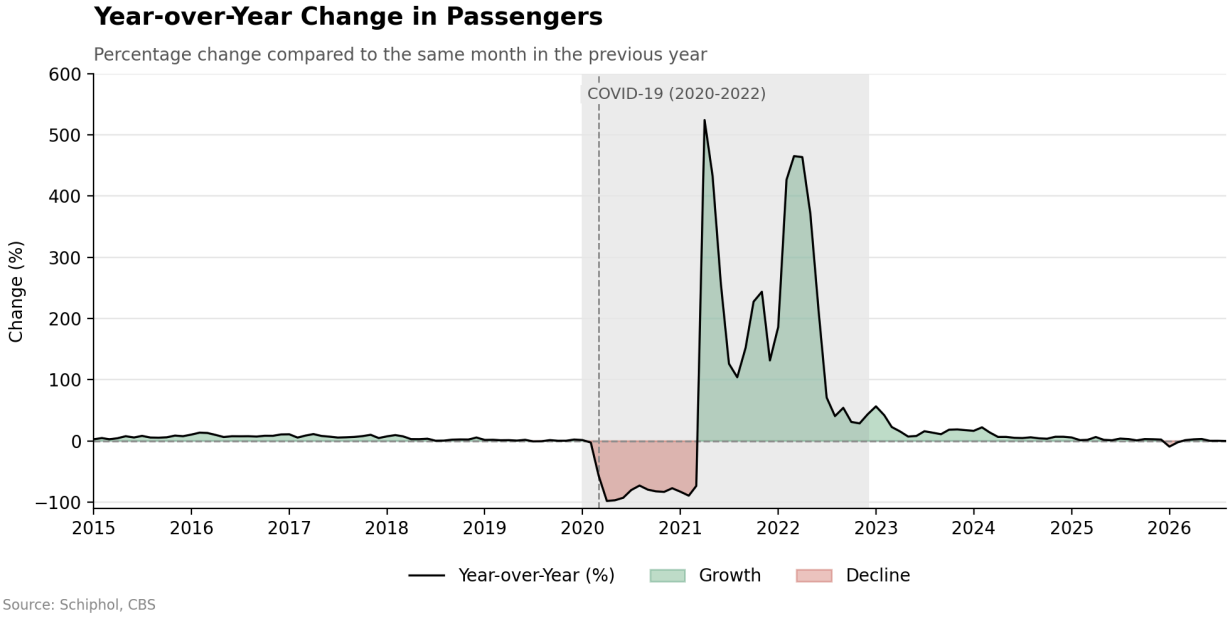
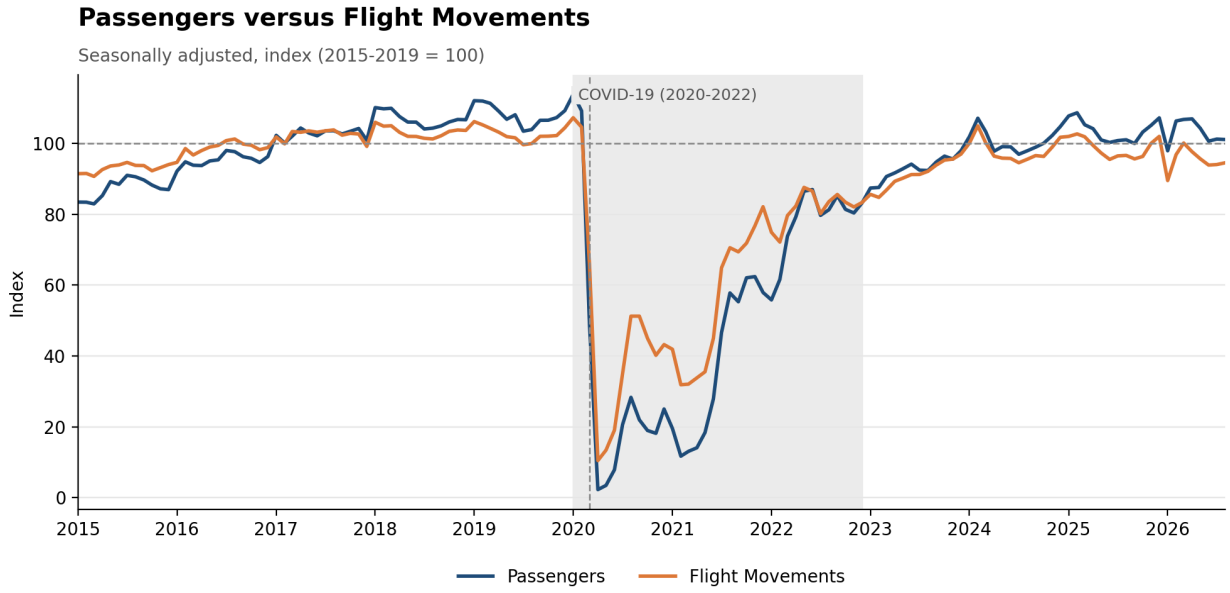


Figure 3. Percentage change compared with the same month one year earlier.

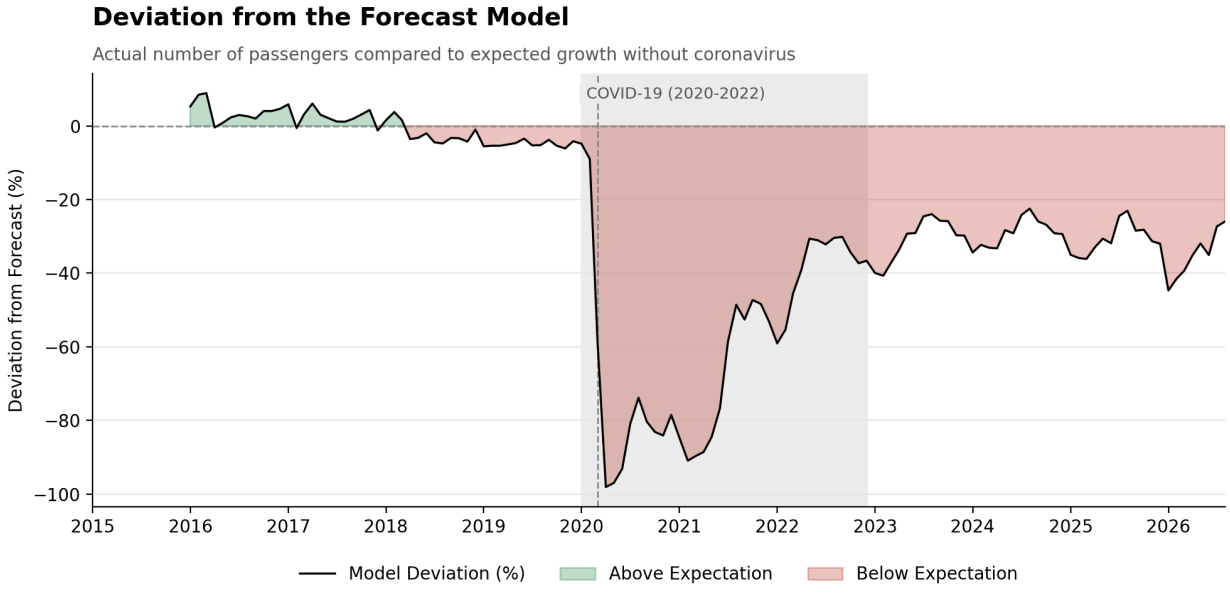
Before 2020, every month grew by between 0 and 14% compared with the year before. From April 2020 to March 2021 every month was 73-98% below the year before, followed by extreme growth figures in 2021-2022 (up to +524% in April 2021), because those months are compared with an almost empty 2020. Since 2024, year-on-year growth is back to normal values, mostly between 0 and 6%.



Source: Schiphol, CBS

Figure 4. Passengers and flight movements, both as a seasonally adjusted index (2015-2019 = 100).

Flight movements fell less deeply than passengers: to an index of 10.4 against 2.3 in April 2020, and in 2020-2021 the flights index stayed well above the passenger index. So planes kept flying, but with far fewer passengers on board. From 2022 the two lines move together again; since 2025 the passenger index is slightly above the flights index, so there are more passengers per flight than in 2015-2019.



Source: Schiphol, CBS

Figure 5. Actual passengers compared with a forecast that continues the average growth of the five years before 2020.

If Schiphol had kept growing as it did before COVID-19, passenger numbers would now be about 25-35% higher than they actually are. Part of this gap already existed before COVID-19: in 2018-2019 growth slowed down, so the model was already slightly too optimistic then. The gap therefore shows the missed growth, not only the effect of COVID-19 itself.

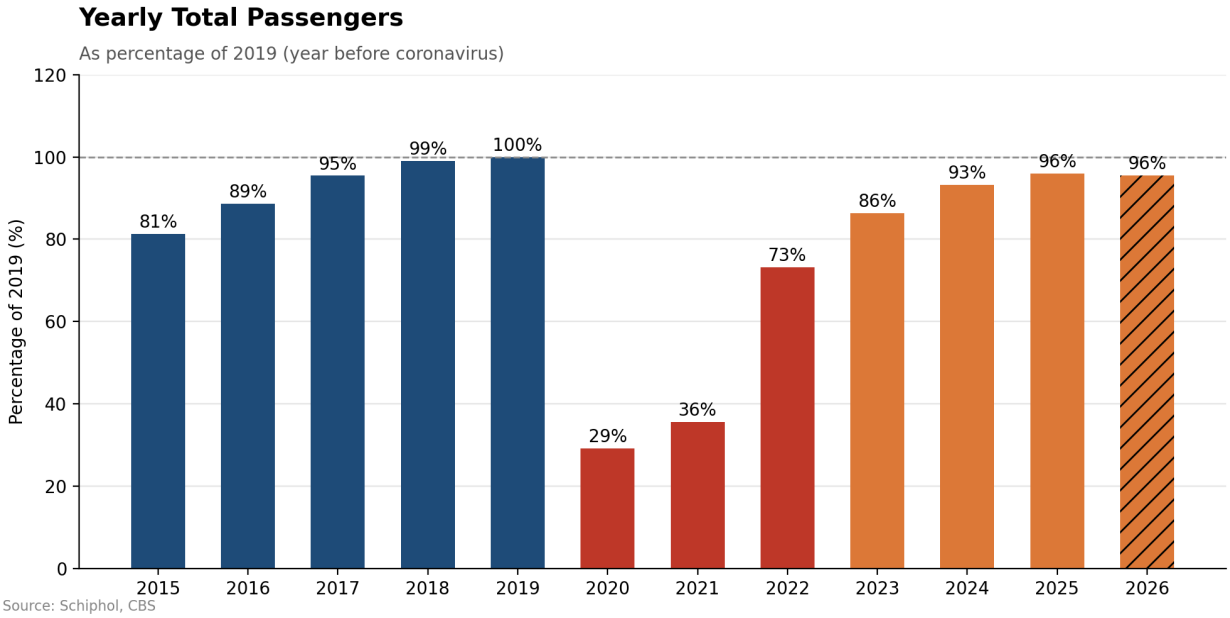


Figure 6. The hatched year 2026 runs through August and is compared with the same months in 2019.

Over 2020 as a whole, Schiphol reached 29% of its 2019 level, and 36% in 2021. Most of the recovery came in 2022 (73%), and it levelled off afterwards. Against 2019, the busiest year, the recovery is not yet complete: 95.9% in 2025 and 95.5% in 2026 through August.

2. Did Dutch people really go on fewer vacations?

Fewer flights do not by themselves show that people went on fewer vacations, because Schiphol also serves business travellers, transfer passengers and foreign visitors. We therefore compare the flights with the vacation figures from Statistics Netherlands (CBS). These come from a separate CBS survey of residents, not from Schiphol.

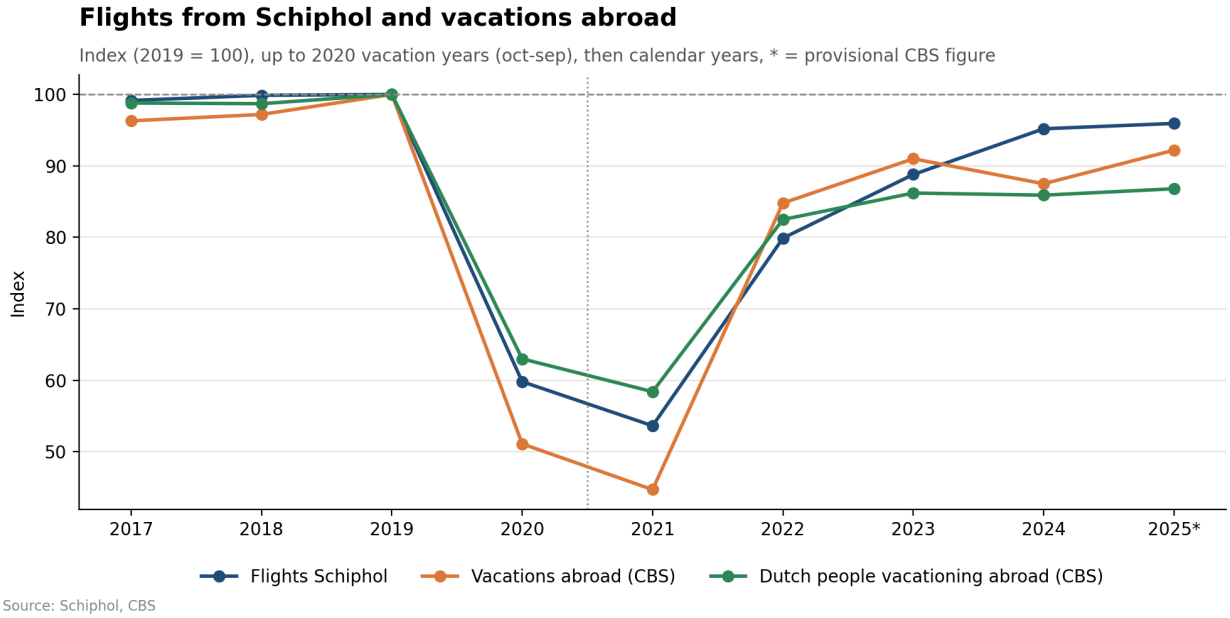


Figure 7. The dotted line marks the switch at CBS from vacation years to calendar years (see section 3).

Year	Flights Schiphol	Vacations abroad	Dutch people vacationing abroad	Vacations per participant
2017	99.2	96.3	98.8	2.01
2018	99.9	97.2	98.7	2.03
2019	100.0	100.0	100.0	2.06
2020	59.8	51.1	63.0	1.67
2021	53.6	44.7	58.4	1.58
2022	79.9	84.8	82.5	2.12
2023	88.8	91.0	86.2	2.18
2024	95.2	87.5	85.9	2.10
2025*	95.9	92.2	86.8	2.19

First three columns: index, 2019 = 100. * Provisional CBS figure.

In 2020 flights fell by 40%, vacations by 49% and the number of Dutch people going abroad by 37%. In 2021 all three stayed below 2020; the recovery only came in 2022. The relationship is therefore strong, with a correlation of 0.977 with vacations and 0.967 with participants. These high values are partly driven by the large shock: without 2020-2021 they are 0.833 and 0.808, based on 7 years.

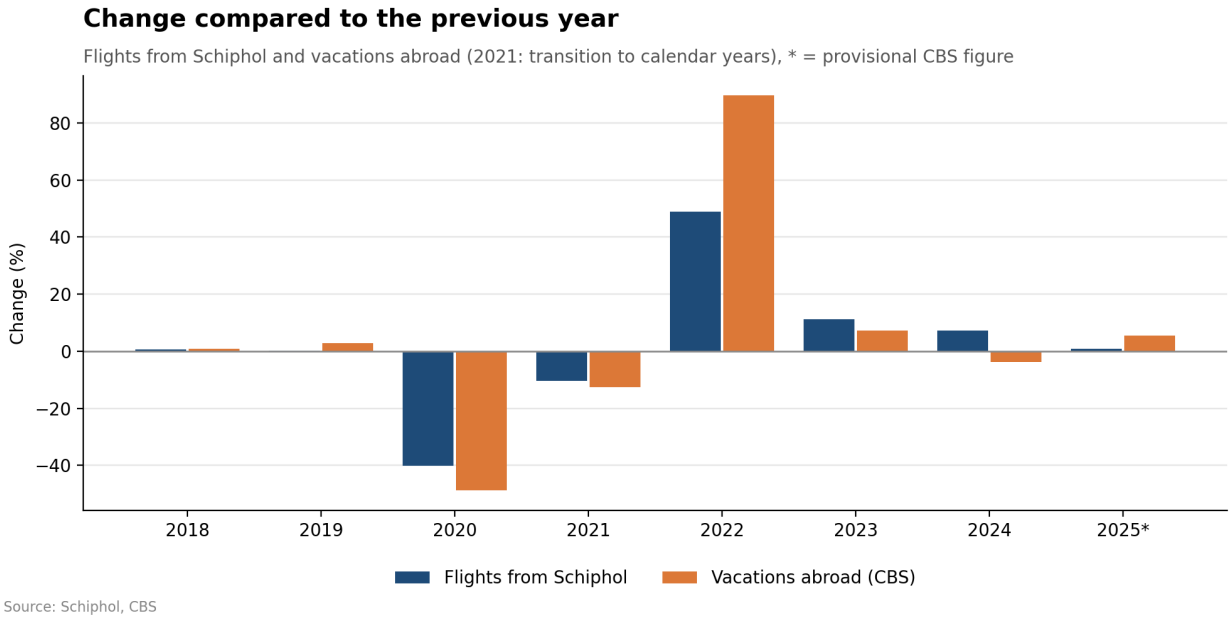


Figure 8. Only in 2024 do flights and vacations clearly diverge: flights +7.2%, vacations -3.9%.

Vacations fell more sharply in 2020 than the number of people. That is because the people who did go, went less often: from 2.06 vacations abroad per participant in 2019 to 1.67 in 2020 and 1.58 in 2021.

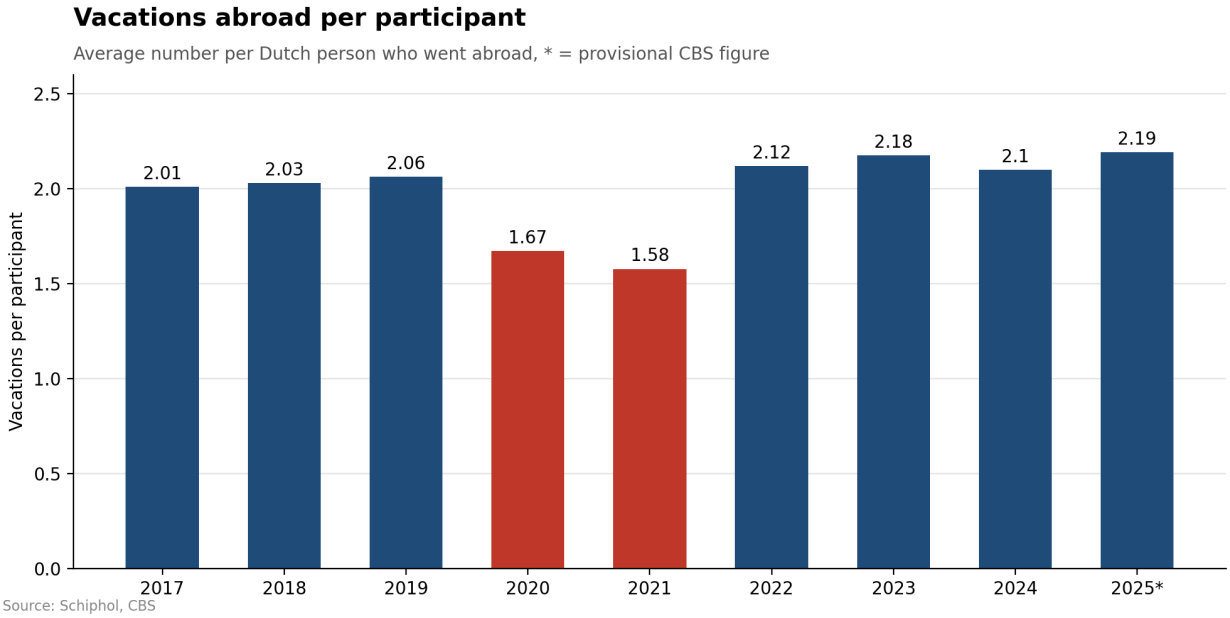


Figure 9. Number of vacations abroad divided by the number of Dutch people who went on vacation abroad.

After COVID-19 the picture is reversed. In 2025 the frequency, at 2.19, is above 2019, while the number of participants remains at 87%. Fewer people are going more often, and flights have recovered further than the number of Dutch vacationers.

3. Data and reliability

- **Schiphol data.** The annual totals differ from CBS by at most 0.0012% (passengers) and 0.0025% (flight movements). This was checked for 2017-2019 and 2022-2024; CBS aviation figures for 2020-2021 are missing from our file.
- **Vacation figures.** All values used for 2017-2025 were checked against CBS tables 84363 (up to 2020) and 84367 (from 2021).
- **Year definition.** Up to 2020, CBS counts in vacation years: 1 October to 30 September, by departure date. From 2021, CBS counts in calendar years, by date of return. The flights are counted the same way, so 2020 here runs from October 2019 to September 2020 and still includes months from before COVID-19. October to December 2020 falls in no CBS year and is left out. Section 1 does use ordinary calendar years.

4. Limitations

- **No causal evidence.** Travel restrictions hit flights and vacations at the same time. The figures show a relationship, not that fewer flights led to fewer vacations.
- **Flights are not only vacation flights.** The data covers all take-offs and landings at Schiphol, regardless of travel purpose. Departing flights are roughly half of them, so the index stays practically the same.
- **Vacations are not only by air.** CBS also counts trips by car and via other airports, for residents aged 15 and over. These figures do not show whether people travelled by car or

closer to home more often during COVID-19.

- **Short series.** There are 9 annual data points, the year definition changes between 2020 and 2021, and 2025 is provisional. Comparisons around that transition are therefore less robust.
- **Spending.** Vacation spending was deliberately not used: CBS reports a trend break for 2025 due to a change in the survey question.

Sources

Schiphol, Traffic and transport figures. CBS, Vakanties van Nederlanders; kerncijfers 2017-2020 (84363NED). CBS, Vakanties van Nederlanders; buitenland, bestemmingen (84367NED). CBS, Maandcijfers Nederlandse luchthavens van nationaal belang (37478hvv). Calculations: results_covid.ipynb.